

Thermodynamic Exergy Analysis of a Turboprop Engine

ENME 637 Thermal Systems Analysis

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December 6, 2017

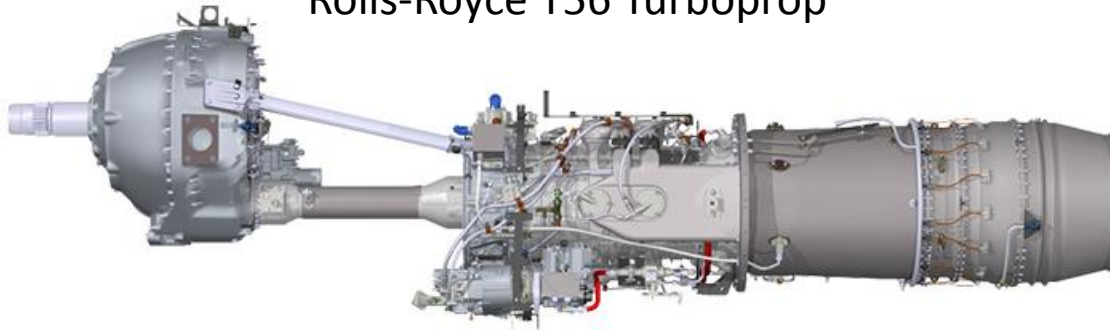
The Turboprop Engine

Turbofan engine



Source: General Electric

Rolls-Royce T56 Turboprop



Source: Rolls-Royce

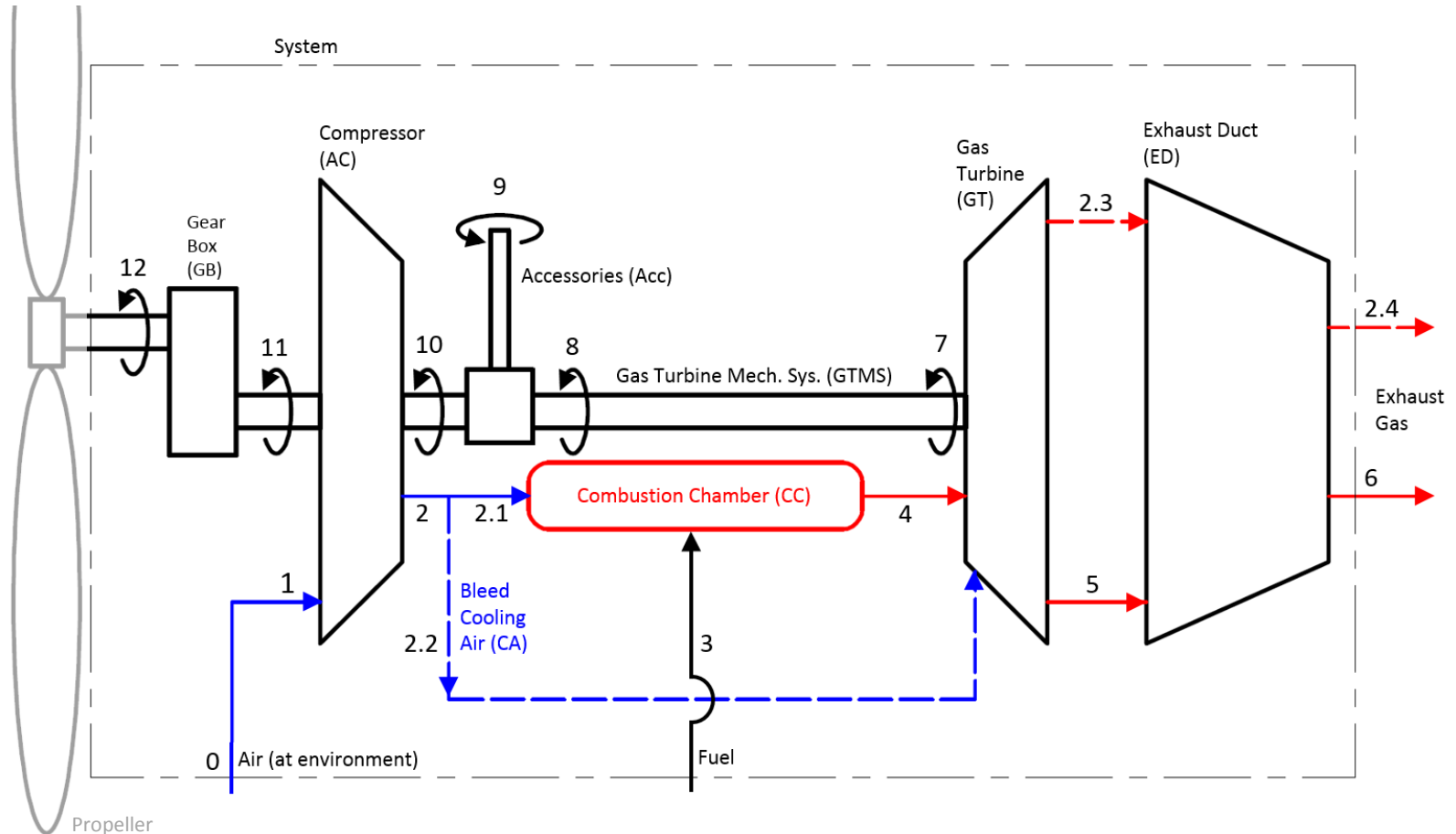
Reciprocating engine



Source: Lycoming

Turboprop Engine Case Study: T56

Basic Components and State Locations

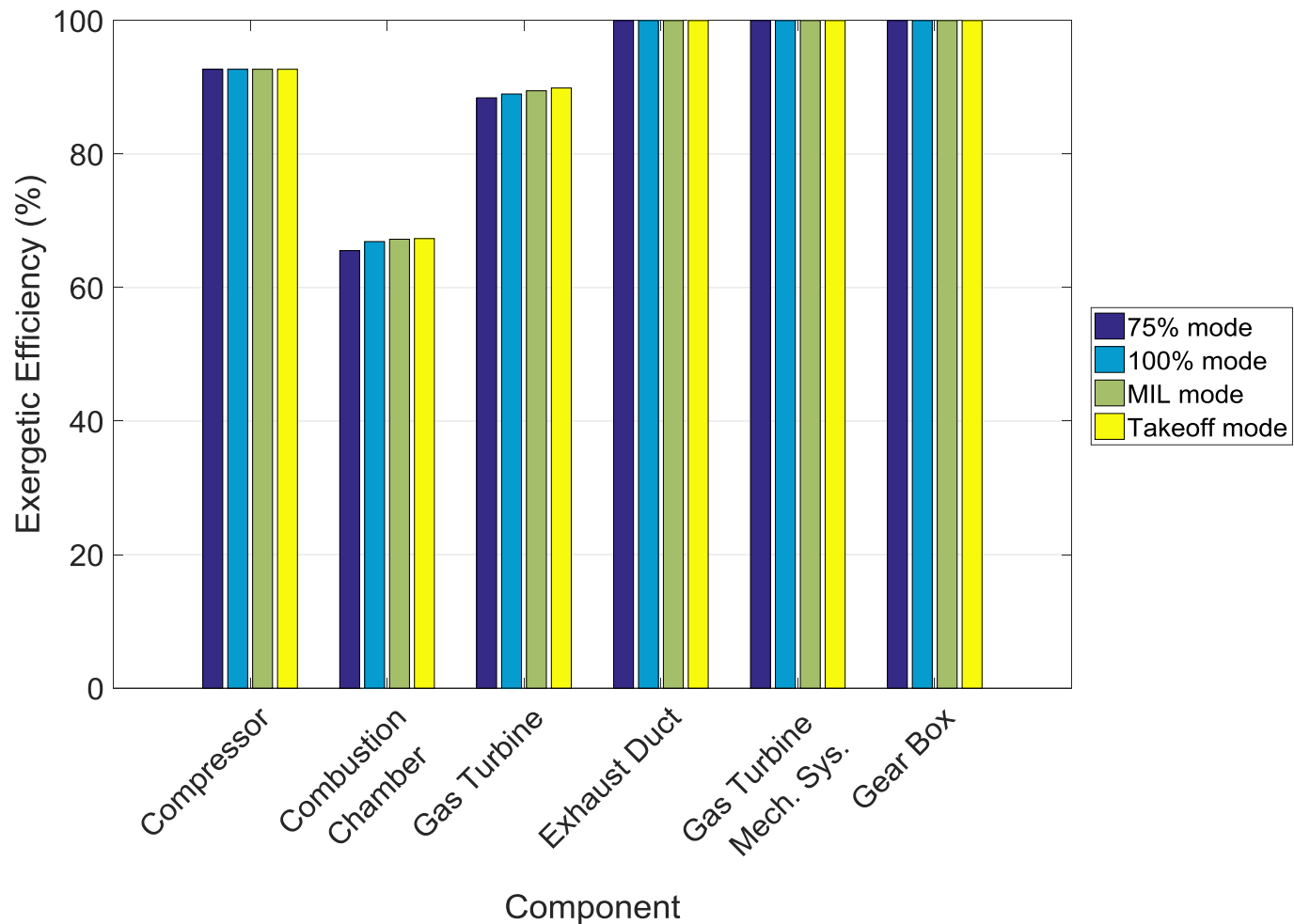


Exergy Analysis Key Assumptions

1. All processes are steady-flow.
2. Air and combustion gas behave as ideal gases.
3. Potential energy (or exergy) differences between the inlet and exit for all processes are negligible.
4. Compressor and combustion chamber kinetic energy (or exergy) differences between the inlet and exit are negligible. **Turbine exit kinetic energy (or exergy) is included.**
5. Compressor, combustion chamber, and turbine processes are adiabatic. **Exhaust duct is internally reversible, not adiabatic.**
6. Fuel is JP-8 jet fuel, chemical formula $C_{12}H_{23}$, LHV 42,800 kJ/kg. For adiabatic flame temperature, Jet-A fuel was used.
7. Air consists of 79% N_2 and 21% O_2 , with complete combustion (and excess air) producing CO_2 , H_2O , O_2 , and N_2 .
8. 10% of compressed air is redirected for turbine blade cooling.

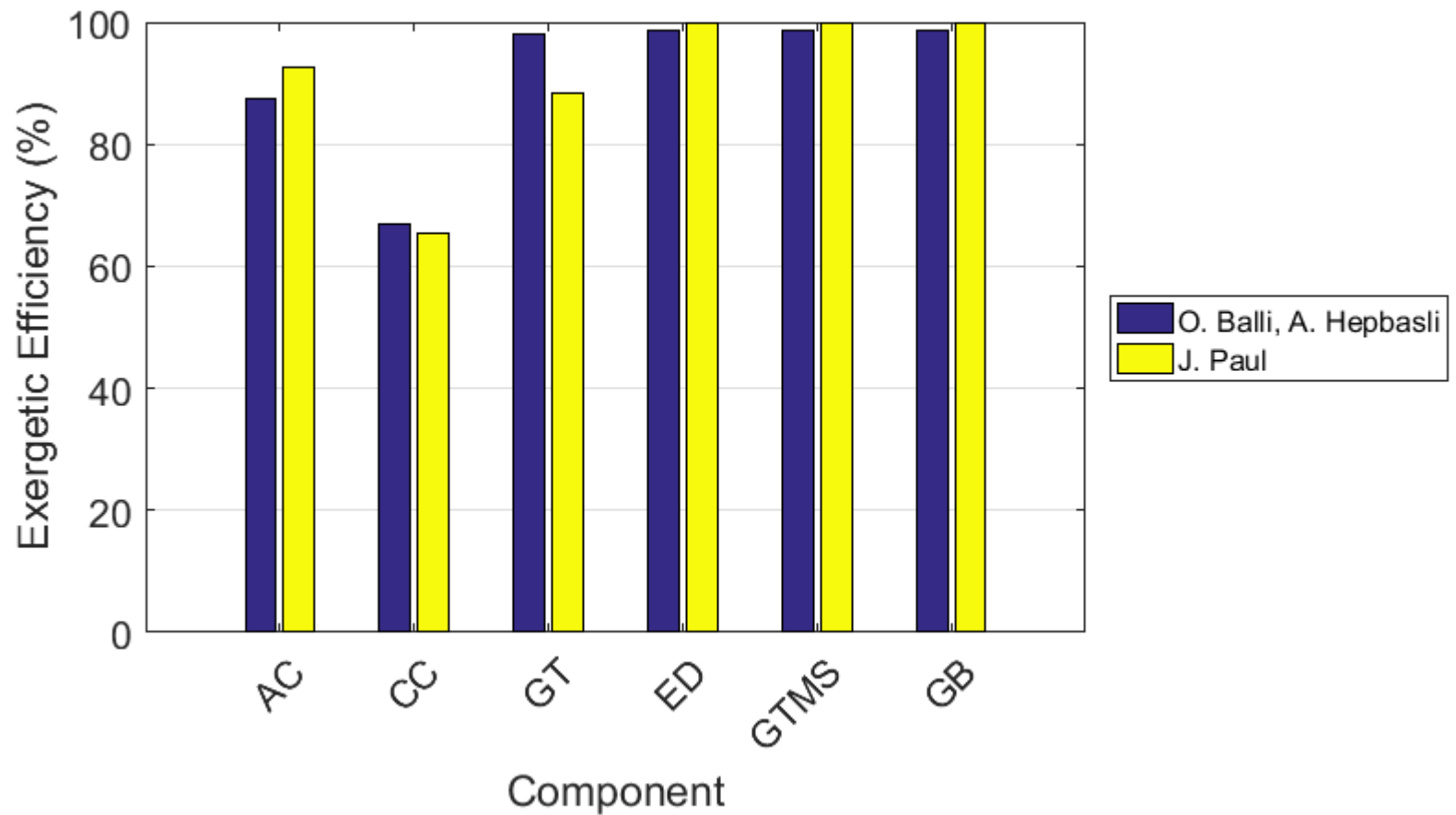
Results

Component Exergetic Efficiency for Four Operation Modes



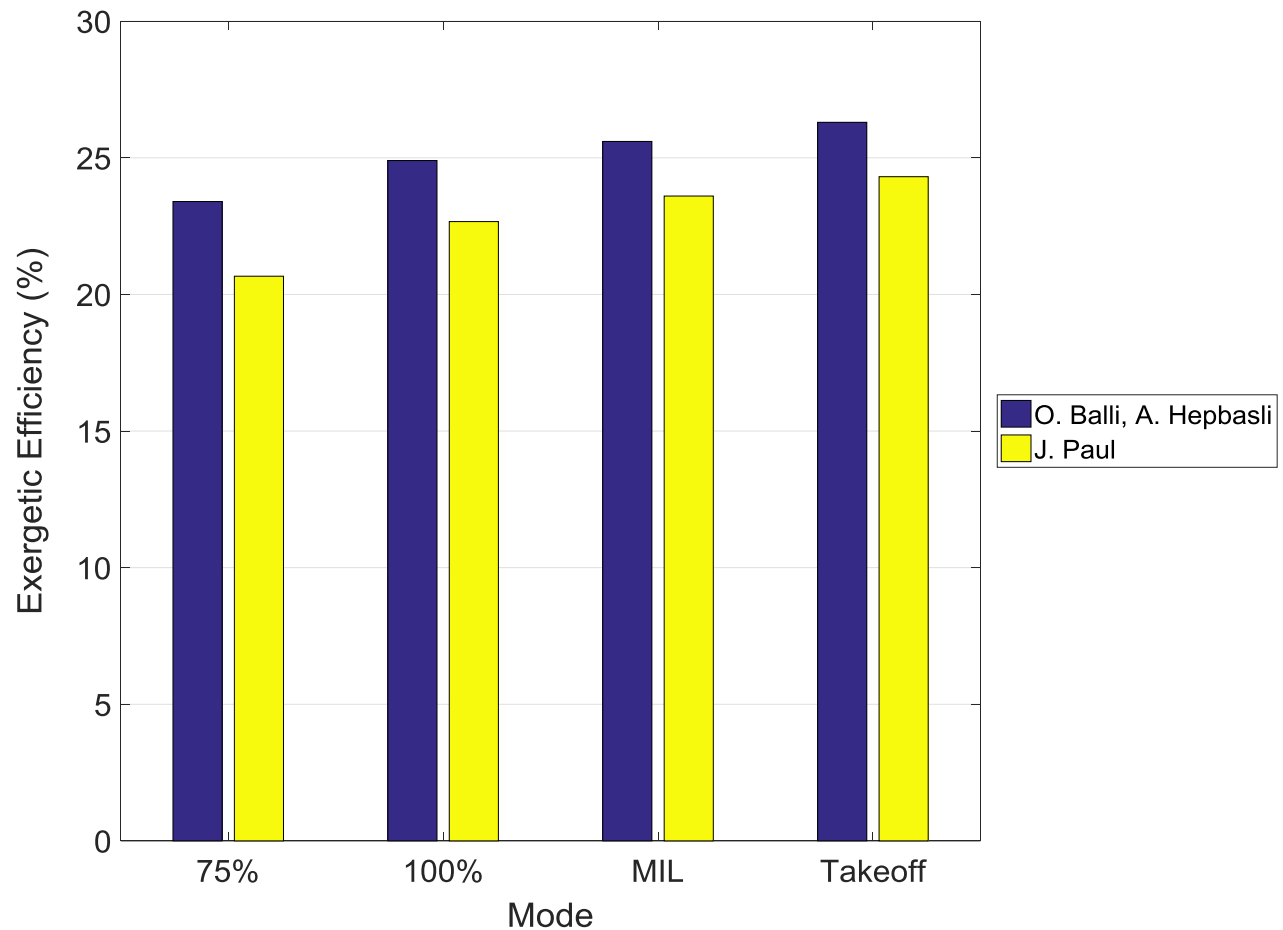
Results

Comparison of Component Exergetic Efficiency for 75%-Mode



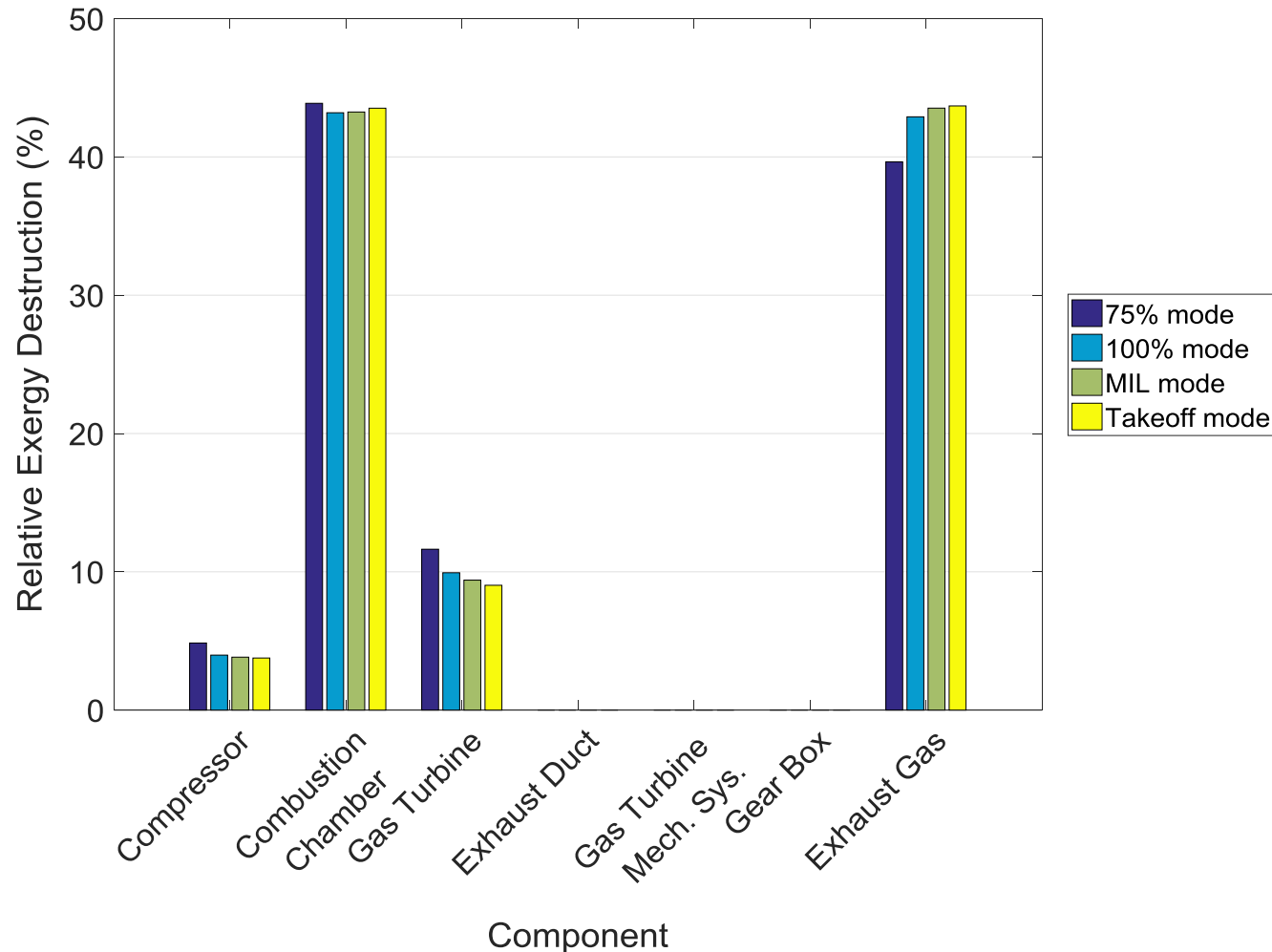
Results

Comparison of Whole Engine Exergetic Efficiency for Four Operation Modes



Improvements to the Turboprop Engine

Relative Exergy Destruction by Component and Mode



Improvements to the Exergy Analysis

- Investigate other methods for estimating chemical exergy of fuel.
- Source of measurement data error (shaft power or flow rates).
- More effort into how to handle the cooling air. (Conservative approach was used.)